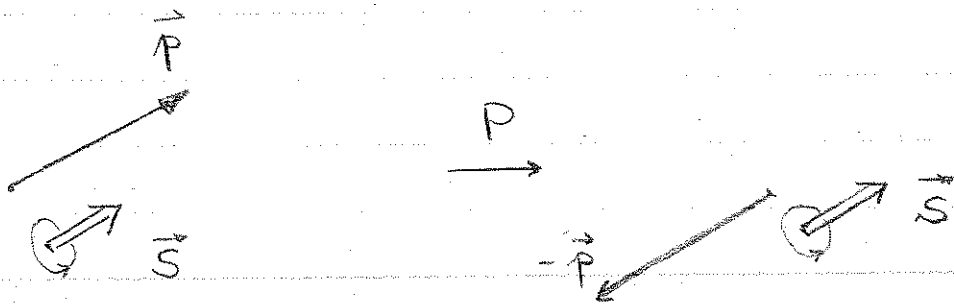


Homework Set # 12

S11.4.3



P = parity $\vec{p}$ = momentum

$\vec{S}$ = spin angular momentum

Under parity $\vec{r} \rightarrow -\vec{r}$, $t \rightarrow t$

$$\vec{p} = m \frac{d\vec{r}}{dt} \rightarrow -\vec{p}$$

$$\vec{L} = m \vec{r} \times \frac{d\vec{r}}{dt} \rightarrow \vec{L}$$

$$\vec{S} \rightarrow \vec{S} \quad (\text{like } \vec{L})$$

If parity is conserved, the state with $\vec{S}$ anti-parallel to $\vec{p}$ is equally likely than the state with $\vec{S}$ parallel to $\vec{p}$ (assuming that the rest of the experiment is parity symmetric).

S11.4.4

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ -z \end{pmatrix}$$

under reflection
in a mirror
located in the
(x, y) plane

$$\begin{pmatrix} x \\ y \\ -z \end{pmatrix} \rightarrow \begin{pmatrix} \cos \theta x + \sin \theta y \\ -\sin \theta x + \cos \theta y \\ -z \end{pmatrix}$$

under
rotation
by angle
 θ around
the z axis

For $\theta = 180^\circ$

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ -z \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix} = -\vec{r}$$

11.4.4. parity $\vec{r} \rightarrow -\vec{r}$: parity transformation

$$\Pi = -I_3$$

$$R_z(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_z(\pi) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_z(\theta) \Pi = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \Pi R_z(\theta)$$

$$R_z(\theta) \Pi \vec{r} = \begin{pmatrix} x \\ y \\ -z \end{pmatrix} \checkmark$$

S12.3.3

$$\psi(r, \varphi) = A e^{-r^2/2\Delta^2} \cos^2 \varphi$$

$$= R(r) \left(e^{2i\varphi} + 2 + e^{-2i\varphi} \right)$$

$$= \sqrt{2\pi} R(r) \left[\psi_{l_z=2k}(\varphi) + 2 \psi_{l_z=0}(\varphi) + \psi_{l_z=-2k}(\varphi) \right]$$

where $\psi_{l_z=mk}(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}$

The probabilities to find the particle in the states of angular momentum

$$l_z = 2k \quad 0 \quad -2k$$

are in the ratios

$$1 : 2^2 : 1$$

The sum of probabilities must be one.

Hence the probabilities are

$$\frac{1}{6}, \quad \frac{4}{6} = \frac{2}{3}, \quad \frac{1}{6}$$

S12.3.6

$$L = \frac{\mu}{2} a^2 \dot{\phi}^2 = L(\dot{\phi})$$

$$p_{\phi} = \frac{\partial L}{\partial \dot{\phi}} = \mu a^2 \dot{\phi} = l_z$$

$$\begin{aligned} H(p_{\phi}) &= \dot{\phi} p_{\phi} - L = \frac{1}{2\mu a^2} p_{\phi}^2 \\ &= \frac{1}{2\mu a^2} l_z^2 \end{aligned}$$

$$H = \frac{1}{2\mu a^2} L_z^2$$

We know that

$$L_z |l_z\rangle = l_z |l_z\rangle$$

requires

$$l_z = m\hbar$$

$$m = 0, \pm 1, \pm 2$$

This

$$H |m\rangle = E_m |m\rangle$$

$$\text{with } |m\rangle = |l_z = m\hbar\rangle, \quad E_m = \frac{m^2 \hbar^2}{2\mu a^2}$$

$E_m = E_{-m}$ because, for a given magnitude of l_z , the energy is the same whether the particle moves clockwise or counterclockwise.

SI 2.3.7 $H = \frac{1}{2\mu} (P_x^2 + P_y^2) + \frac{1}{2} \mu \omega^2 (x^2 + y^2)$

1.

$$[L_z, x^2 + y^2] = 2 [L_z, x] x + 2 [L_z, y] y \\ = 2 i \hbar (y x - x y) = 0$$

Similarly $[L_z, P_x^2 + P_y^2] = 0$

$$[L_z, H] = 0$$

$$H |E, m\rangle = E |E, m\rangle$$

$$m = 0, \pm 1, \pm 2, \dots$$

$$L_z |E, m\rangle = \hbar m |E, m\rangle$$

$$\langle \rho, \phi | E, m \rangle = \psi_{E, m}(\rho, \phi)$$

$$= R_{E, m}(\rho) \frac{1}{\sqrt{2\pi}} e^{i m \phi}$$

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + \frac{1}{2} \mu \omega^2 (x^2 + y^2) \right] \psi_{E, m} = E \psi_{E, m}$$

$$\nabla^2 = \frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2}{\partial \phi^2}$$

$$-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho} \frac{d}{d\rho} - \frac{m^2}{\rho^2} \right) R_{E, m} + \frac{\mu \omega^2}{2} \rho^2 R_{E, m} = E R_{E, m}$$

ii

Let $R_{E,m} \sim \rho^p + o(\rho^{p+1})$ as $\rho \rightarrow 0$

$$= \frac{\hbar^2}{2\mu} [p(p-1) + p - m^2] \rho^{p-2} + o(\rho^{p-1})$$

$$+ \frac{\mu\omega^2}{2} \rho^{p+2} + o(\rho^{p+1}) = E \rho^p + o(\rho^{p+1})$$

$$\therefore \rho^2 - p + p - m^2 = 0$$

$$\therefore \rho^2 = m^2 \quad \therefore \rho = \pm m$$

For $m=0$ $R_{E,m} \sim \text{constant}$ as $\rho \rightarrow 0$

For $m = \pm 1, \pm 2, \dots$

$$R_{E,m} \sim \rho^{\pm|m|}$$

However, the wavefunction must be normalizable

$$\therefore \int_0^\infty \rho d\rho |R_{E,m}|^2 < \infty$$

$$\therefore \rho > -1 \quad (\rho = -1, -2, \dots \text{ are excluded})$$

$$\therefore R_{E,m} \sim \rho^{|m|} \quad \text{for any } m.$$

III

For $p \rightarrow \infty$, ER_{Em} and $\frac{\hbar^2 m^2}{2\mu p^2} R_{Em}$

are negligible compared to $\frac{\mu\omega^2}{2} p^2 R_{Em}$

Thus

$$R_{Em}(p) \sim R_E^\infty(p) \quad \text{as } p \rightarrow \infty$$

with

$$-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dp^2} + \frac{1}{p} \frac{d}{dp} \right) R_E^\infty + \frac{\mu\omega^2}{2} p^2 R_E^\infty = 0$$

Let $R_E^\infty = e^\varphi$

$$-\frac{\hbar^2}{2\mu} \left(\varphi'' + \varphi'^2 + \frac{1}{p} \varphi' \right) + \frac{\mu\omega^2}{2} p^2 = 0$$

We need $\varphi \rightarrow -\infty$ as $p \rightarrow \infty$
in order that

$$\int_0^\infty p dp |R_E^\infty|^2 < \infty$$

Therefore let $\varphi \sim -c p^p$ $c > 0$
 $p > 0$

$$\varphi'' \sim p^{p-2}, \quad \varphi'^2 \sim p^{2(p-1)}$$

$$\frac{1}{p} \varphi' \sim p^{p-2}$$

Hence φ'^2 dominates

$$+ \frac{\hbar^2}{2\mu} \varphi'^2 = \frac{\mu\omega^2}{2} \rho^2 \quad \text{as } \rho \rightarrow \infty$$

$$\varphi = - \frac{\mu\omega}{\hbar} \frac{1}{2} \rho^2$$

($R \sim e^{+\frac{1}{2} \frac{\mu\omega}{\hbar} \rho^2}$ is not allowed)

$$R_E^\infty = e^{-\frac{\mu\omega}{\hbar} \frac{1}{2} \rho^2}$$

iv $\epsilon = \frac{E}{\hbar\omega} \quad y = \left(\frac{\mu\omega}{\hbar}\right)^{1/2} \rho$

$$-\frac{1}{2} \left(\frac{d^2}{dy^2} + \frac{1}{y} \frac{d}{dy} - \frac{m^2}{y^2} \right) R + \frac{1}{2} y^2 R = \epsilon R$$

v $R_{E,m} = \Omega e^{-\frac{1}{2} y^2} = U y^{|m|} e^{-\frac{1}{2} y^2}$

$$R' = (-y\Omega + \Omega') e^{-\frac{1}{2} y^2}$$

$$R'' = [(-1 + y^2)\Omega - 2y\Omega' + \Omega''] e^{-\frac{1}{2} y^2}$$

$$-\frac{1}{2} [(-1 + y^2)\Omega - 2y\Omega' + \Omega'' - \Omega + \frac{1}{y}\Omega'] - \frac{m^2}{y^2}\Omega + \frac{1}{2} y^2 \Omega = \epsilon \Omega$$

$$+ \Omega'' + \left(\frac{1}{y} - 2y \right) \Omega' - \left(2 + \frac{m^2}{y^2} \right) \Omega = -2E \Omega$$

$$\Omega' = y^{|m|} \left(\frac{|m|}{y} U + U' \right)$$

$$\Omega'' = y^{|m|} \left[\frac{m^2 - |m|}{y^2} U + \frac{2|m|}{y} U' + U'' \right]$$

$$U'' + U' \left(\frac{2|m|}{y} + \frac{1}{y} - 2y \right)$$

$$+ U \left(\frac{m^2 - |m|}{y^2} + \left(\frac{1}{y} - 2y \right) \frac{|m|}{y} - 2 - \frac{m^2}{y^2} \right)$$

$$= -2E U$$

$$U'' + \left(\frac{2|m|+1}{y} - 2y \right) U' + (2E - 2 - 2|m|) U = 0$$

$$\underline{\text{vi.}} \quad \begin{array}{ccccccc} -2 & & -2 & & 0 & & 0 \end{array}$$

The differential operator on the LHS of the equation shifts a power of y either by -2 or 0 .

VII

$$U = \sum_{r=0}^{\infty} C_r y^r$$

$$\sum_{r=1}^{\infty} C_r y^{r-2} [r(r-1) + 2|m|+1]$$

$$+ \sum_{r=0}^{\infty} C_r y^r [-2r + 2\epsilon - 2 - 2|m|] = 0$$

$$C_{r+2} = -C_r \frac{2[\epsilon - 1 - |m| - r]}{(r+2)(r+1) + 2|m|+1}$$

$C_1 = 0$ because the first term in the first sum

$$C_1 \frac{1}{y} (2|m|+1)$$

must vanish by itself

$$\text{Hence } C_3 = C_5 = C_7 = \dots = 0$$

The series is even

If the series does not terminate

($\epsilon \neq 1 + |m| + r$ for all even r),

$$C_{r+2} \sim \frac{2}{r+2} C_r \quad \text{for large } r, \text{ and}$$

$$U \sim \sum_{p=0}^{\infty} \frac{1}{p!} y^{2p} = e^{y^2}$$

which is the rejected solution raising its ugly head.

We need to have

$$E = 1 + |m| + 2k \quad \text{for some positive integer } k$$

$$E = (2k + |m| + 1) \hbar \omega$$

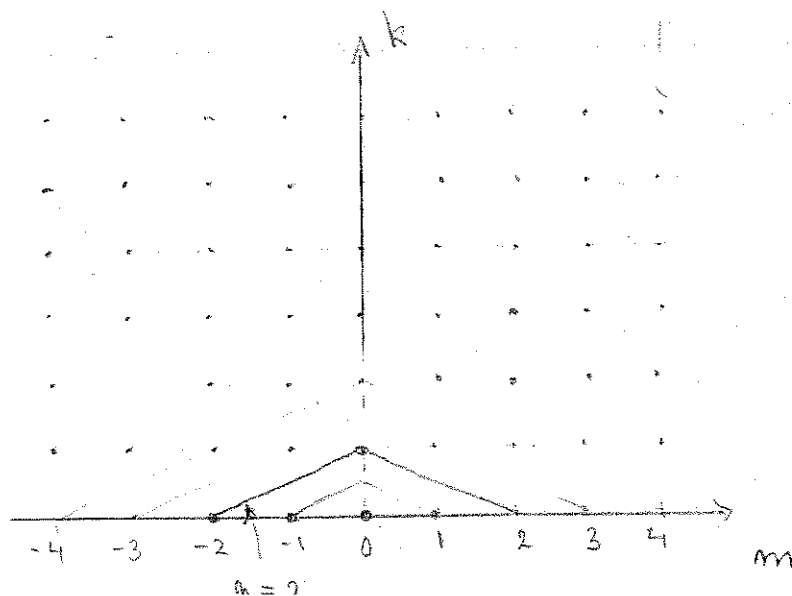
$$= (n + 1) \hbar \omega$$

with $n = |m| + 2k$

$$m = 0, \pm 1, \pm 2, \dots$$

$$k = 0, 1, 2, 3, \dots$$

VIII



The degeneracy of the n^{th} level
is $n+1$ check:

- for n even : $(k, m) = (0, \pm n)$
 $= (1, \pm(n-2))$
 $\vdots$
 $= (\frac{n}{2} - 1, \pm 2)$
 $= (\frac{1}{2}n, 0)$

$$\text{degeneracy} = \frac{n}{2} \cdot 2 + 1 = n+1$$

- for n odd : $(k, m) = (0, \pm n)$
 $= (1, \pm(n-2))$
 $\vdots$
 $= (\frac{n-1}{2}, \pm 1)$

$$\text{degeneracy} = \frac{n+1}{2} \cdot 2 = n+1$$

In Cartesian coordinates

$$E = \hbar \omega \left(n_1 + \frac{1}{2} + n_2 + \frac{1}{2} \right)$$
$$= \hbar \omega (n+1)$$

$$n = n_1 + n_2$$

$$n_1, n_2 = 0, +1, +2, \dots$$

For given m , $(n_1, n_2) = (0, n), (1, n-1), \dots, (n, 0)$

∴ the degeneracy of the n^{th} level is $n+1$

ix

$$m=0$$

$$m=0$$

$$E = \hbar \omega$$

$$U'' + \left(\frac{1}{y} - 2y \right) U' = 0$$

$$U = \text{constant}$$

or

$$U = e^{y^2}$$

(reject)



$$\psi_{E=\hbar\omega} = N e^{-\frac{\mu\omega}{2\hbar} \rho^2}$$

$$|N|^2 \int_0^\infty \rho d\rho \int_0^{2\pi} d\phi e^{-\frac{\mu\omega}{\hbar} \rho^2}$$

$$= |N|^2 2\pi \frac{\hbar}{2\mu\omega} = 1$$

$$\psi_{E=\hbar\omega} = \sqrt{\frac{\mu\omega}{\pi\hbar}} e^{-\frac{\mu\omega}{2\hbar} \rho^2}$$

$$m = 1$$

$$m = \pm 1$$

$$E = 2\hbar\omega$$

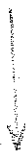
$$U'' + \left(\frac{3}{y} - 2y\right) U' = 0$$

$$U = \text{constant}$$

or

$$U' = \frac{1}{y^3} e^{y^2}$$

$$U \sim e^{y^2} \text{ (reject)}$$



$$\psi_{E=2\hbar\omega, m=\pm 1} = N e^{\pm i\phi} \rho e^{-\frac{\mu\omega}{2\hbar}\rho^2}$$

$$|N|^2 \int_0^\infty \rho d\rho \int_0^{2\pi} d\phi \rho^2 e^{-\frac{\mu\omega}{\hbar}\rho^2}$$

$$= |N|^2 2\pi \left(-\frac{d}{d\alpha}\right) \frac{1}{2\alpha} \Big|_{\alpha = \frac{\mu\omega}{\hbar}}$$

$$= |N|^2 \left(\frac{\hbar}{\mu\omega}\right)^2 \pi = 1$$

$$\psi_{E=2\hbar\omega, m=\pm 1} = \left(\frac{\mu\omega}{\hbar}\right)^{\frac{1}{2}} \frac{1}{\sqrt{\pi}} \rho e^{\pm i\phi} e^{-\frac{\mu\omega}{2\hbar}\rho^2}$$

x. The $E = \hbar\omega$ state is unique.

Therefore

$$\Psi_{E=\hbar\omega}(\rho, \varphi) = \Psi_0(x) \cdot \Psi_0(y)$$

$$= \left(\frac{\mu\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{\mu\omega}{2\hbar}(x^2+y^2)}$$

✓

The two $E = 2\hbar\omega$ states are linear combinations of

$$\Psi_1(x) \Psi_0(y) = \left(\frac{\mu\omega}{\pi\hbar}\right)^{1/2} \frac{1}{\sqrt{2}} e^{-\frac{\mu\omega}{2\hbar}(x^2+y^2)} \frac{1}{2} \left(\frac{\mu\omega}{\hbar}\right)^{1/2} x$$

$$= \left(\frac{\mu\omega}{\hbar}\right) \sqrt{\frac{2}{\pi}} \rho e^{-\frac{\mu\omega}{2\hbar}\rho^2} \cos\varphi$$

and

$$\Psi_0(x) \Psi_1(y) = \left(\frac{\mu\omega}{\hbar}\right) \sqrt{\frac{2}{\pi}} \rho e^{-\frac{\mu\omega}{2\hbar}\rho^2} \sin\varphi$$

We have

$$|n=1, m=\pm 1\rangle = \frac{1}{\sqrt{2}} \left(|n_1=1, n_2=0\rangle \pm i |n_1=0, n_2=1\rangle \right)$$

Under parity

$$\rho \rightarrow \rho$$

$$\varphi \rightarrow \varphi + \pi$$

$$e^{im\varphi} \rightarrow e^{im\pi} e^{im\varphi} = (-1)^m e^{im\varphi}$$

$$\psi_{E,m}(\rho, \varphi) = N R_{E,m}(\rho) e^{im\varphi}$$

$$\rightarrow (-1)^m \psi_{E,m}(\rho, \varphi)$$

But $(-1)^m = (-1)^{n-2k} = (-1)^n$

Thus

$$\psi_{E,m}(\rho, \varphi) \rightarrow (-1)^n \psi_{E,m}(\rho, \varphi)$$

